



Certain DNA vaccines have been tested on cattle.

types of responses to combat infectious microbes: An antibody response and a cell-mediated response. Typically, vaccines activate only the antibody response to an infectious microbe.

Advantages of DNA Vaccines

DNA vaccines represent a new approach to immunization, in that an individual is directly inoculated (injected) with DNA that genetically encodes one or more of the antigens associated with the infectious agent. In effect, the recipient of a DNA vaccine produces the immunizing protein (antigen) within his own cells as a result of the immunization process.

This revolutionary approach to vaccination offers many advantages over conventional vaccines. A major advantage is that DNA vaccination stimulates both the antibody and cell-mediated components of the immune system, whereas conventional protein vaccines usually stimulate only the antibody response. Furthermore, DNA vaccines are simpler to produce and store than conventional vaccines, and are therefore less expensive. Preliminary studies to date indicate that DNA vaccines appear to be very safe and to produce no side effects.

DNA Vaccination Techniques

DNA vaccination involves immunization with a circular piece of DNA, known as a plasmid, that contains the gene (or genes) that code for an

antigen. When injected into an individual, the plasmid is taken up by cells and its genetic information is translated into the immunizing protein. This enables the host immune system to respond to the antigen as it is presented to other cells.

In many respects, this process is reminiscent of what occurs during a viral infection, when viral proteins are expressed within host cells. Thus, a DNA vaccine is somewhat like a very simple, nonreplicating virus. However, plasmid DNA vaccines do not replicate within the host, and therefore do not infect neighboring cells, as occurs during a viral infection.

This innovation in vaccination strategy was discovered some years ago, but the active development of this technology only began after Stephen Johnston's group at the University of Texas Southwestern Medical Center demonstrated that **plasmid** DNA can induce the formation of antibodies against an encoded protein in 1992. Johnston's group was able to show that when mice are inoculated with plasmid DNA encoding human growth hormone, the mice produce antibodies against the hormone.

Shortly thereafter, another research group reported that a protective cell-mediated immune response against influenza virus followed immunization with plasmid DNA encoding an influenza virus protein. This study demonstrated that DNA-based immunization stimulates both components of the immune system and helped to establish that DNA immunization is capable of inducing a protective response against infection.

There are two basic ways to inoculate with plasmid-based vaccines. The first involves direct inoculation into muscle tissue, with the plasmid DNA suspended in a saline (salt) solution ("naked" DNA). The DNA is eventually taken up into nearby cells and processed to express the encoded antigen. The other method uses a high-pressure device, a so-called gene gun, to propel DNA-coated gold particles into cells in the skin. This method is sometimes referred to as **biolistic** particle inoculation. Both methods are widely used, and newer methods for the delivery of plasmid DNA vaccines are currently in development.

As of December 2001 there are several clinical studies in progress to evaluate the effectiveness of DNA vaccination. Most of these studies were targeted against viral infectious agents, such as HIV, hepatitis B, and influenza virus. However, there are also studies in progress to develop DNA vaccines against malaria and tuberculosis. There are even several efforts to develop DNA vaccines against various forms of cancer, an approach which seems to offer significant hope for the future. SEE ALSO IMMUNE SYSTEM GENETICS; PLASMID.

Darrell R. Galloway

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antigen a foreign substance that provokes an immune response

plasmid a small ring of DNA found in many bacteria

biolistic firing a microscopic pellet into a biological sample (from biological/ballistic)



Dominance *See Inheritance Patterns*

Down Syndrome

trisomy presence of three, instead of two, copies of a particular chromosome

Down syndrome, also called **trisomy 21**, is the single most common genetic cause of moderate mental retardation. It occurs in about one of every eight hundred live births. It is caused by the inheritance of an extra copy of chromosome 21. The condition was named after an English physician, J. Langdon Down, who in 1866 published the first report describing patients with similar facial features and mental retardation. The chromosomal basis of Down syndrome was not determined until nearly a century later.

Clinical Features

Down syndrome is associated with a characteristic physical appearance, mental retardation, and specific birth defects or health conditions. The facial features, in addition to low muscle tone (called hypotonia), are often the first signs that alert a physician to a potential diagnosis of Down syndrome. These features include an up-slant of the outer corners of the eyes, small skin folds over the inner corners of the eyes, a small nose with a flat nasal bridge, a flat profile, and a large, grooved tongue that often protrudes from the mouth. Other physical characteristics can include a short neck, excess skin on the back of the neck, short hands with a single palmar crease, a wide gap between the first and second toes, and short stature. There are many individuals without Down syndrome who may have one or more of these features. It is only when the features occur together and the appropriate genetic test (chromosome studies) confirms clinical suspicion that a diagnosis of Down syndrome is made.

All individuals with Down syndrome have mental retardation, usually mild to moderate. The degree of learning disability may not be immediately apparent, since social ability generally exceeds scholastic ability. Early intervention programs are important for giving people with Down syndrome the best chance to maximize their learning potential.

Certain birth defects and health conditions are more common in individuals with Down syndrome. The most common birth defect is a congenital heart defect, affecting 40 percent to 50 percent of newborns with the condition. Although many can be repaired with surgery, congenital heart defects remain the major cause of early death among affected persons. Individuals with Down syndrome have an increased chance of experiencing hearing loss, vision problems, and thyroid disease, as well as an increased susceptibility to infections. Because of such concerns, specific guidelines for the health care of individuals with Down syndrome have been developed.

Chromosomal Basis of Down Syndrome

In 1959 French geneticist Jerome Lejeune recognized that individuals with Down syndrome have forty-seven chromosomes instead of the usual forty-six. Later, it was determined that it is an extra copy of chromosome 21 that causes the condition. It is not yet clear how the extra chromosome causes the clinical features, although it is believed that an “extra dose” of one or more of the genes on the chromosome is responsible.



Down syndrome is a chromosome disorder that causes mental retardation. Characteristic features of Down syndrome include a flattened midface, small ears and mouth, and short, broad hands.

There are three types of Down syndrome: trisomy 21, mosaic Down syndrome, and translocation Down syndrome. In 94 percent of cases, the extra copy of chromosome 21 stands alone (is not attached to any other chromosomes) and is present in every cell of the body. This is called trisomy 21, trisomy meaning three.

Trisomy 21 occurs due to a chromosome packaging error. Usually when the body makes its sex cells (egg or sperm cells) during **meiosis**, it packages up one chromosome from each pair. However, sometimes an error (nondisjunction) occurs, causing both chromosomes from a pair to get packaged together. If the sex cell with the extra chromosome is fertilized by a sex cell with the usual chromosome number, the resulting embryo will have a trisomy. If the extra chromosome is chromosome 21, the embryo will have Down syndrome. About 75 percent of embryos with trisomy 21 abort spontaneously before birth. Nondisjunction occurs by chance in the making of both egg and sperm cells, but it happens more often in egg cells as women get older. Thus, the chance of having a baby with Down syndrome increases with increasing maternal age.

meiosis cell division that forms eggs or sperm

Translocation Down syndrome, which accounts for 3 percent to 4 percent of cases, occurs when the extra copy of chromosome 21 is attached to another chromosome. In about one-fourth of the cases where a person has translocation Down syndrome, he or she inherited the translocation from a parent. Therefore it is important to test the parents' chromosomes in these cases, for purposes of future family planning.

The third type of Down syndrome is the mosaic type, which occurs in 2 percent to 3 percent of cases. In mosaic Down syndrome, a person has some cells with an extra copy of chromosome 21 and some cells with the usual two copies. People with mosaic Down syndrome may or may not have milder symptoms than people with "full" trisomy 21.

Testing for Down Syndrome

Cytogenetic analysis looks at the number and structure of a person's chromosomes. This test, which can be performed on a blood sample, is the test used to definitively determine if an individual has Down syndrome.



Risk of Down syndrome and other chromosomal abnormalities increases with maternal age. Adapted from Evans, 1989.

Maternal Age	Risk of Down Syndrome	Total Risk for all Chromosomal Abnormalities
20	1/1667	1/526
21	1/1667	1/526
22	1/1429	1/500
23	1/1429	1/500
24	1/1250	1/476
25	1/1250	1/476
26	1/1176	1/476
27	1/1111	1/455
28	1/1053	1/435
29	1/1000	1/417
30	1/952	1/385
31	1/909	1/385
32	1/769	1/322
33	1/602	1/286
34	1/485	1/238
35	1/378	1/192
36	1/289	1/156
37	1/224	1/127
38	1/173	1/102
39	1/136	1/83
40	1/106	1/66
41	1/82	1/53
42	1/63	1/42
43	1/49	1/33
44	1/38	1/26
45	1/38	1/21
46	1/23	1/16
47	1/18	1/13
48	1/14	1/10
49	1/11	1/8

amniocentesis removal of fluid from the amniotic sac surrounding a fetus, for diagnosis

Prenatal diagnosis for Down syndrome (testing for the condition during pregnancy) is possible. Chromosome studies can be performed on fetal cells collected via chorionic villus sampling (CVS) at ten to twelve weeks of pregnancy or by **amniocentesis** at fifteen to twenty weeks of pregnancy. Because of the link between the mother's age and the chance of Down syndrome, prenatal diagnosis for Down syndrome and other chromosome conditions is routinely offered to women thirty-five and older. Whether to pursue prenatal diagnosis is a personal decision that can only be made by the parents. SEE ALSO BIRTH DEFECTS; CHROMOSOMAL ABERRATIONS; MEIOSIS; MOSAICISM; NONDISJUNCTION; PRENATAL DIAGNOSIS.

Angela Trepanier and Gerald L. Feldman

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anabolic steroids hormones used to build muscle mass

- anaerobic** without oxygen or not requiring oxygen
- androgen** testosterone or other masculinizing hormone
- anemia** lack of oxygen-carrying capacity in the blood
- aneuploidy** abnormal chromosome numbers
- angiogenesis** growth of new blood vessels
- anion** negatively charged ion
- anneal** join together
- anode** positive pole
- anterior** front
- antibody** immune-system protein that binds to foreign molecules
- antidiuretic** a substance that prevents water loss
- antigen** a foreign substance that provokes an immune response
- antigenicity** ability to provoke an immune response
- apoptosis** programmed cell death
- Archaea** one of three domains of life, a type of cell without a nucleus
- archaeans** members of one of three domains of life, have types of cells without a nucleus
- aspirated** removed with a needle and syringe
- aspiration** inhalation of fluid or solids into the lungs
- association analysis** estimation of the relationship between alleles or genotypes and disease
- asymptomatic** without symptoms
- ATP** adenosine triphosphate, a high-energy compound used to power cell processes
- ATPase** an enzyme that breaks down ATP, releasing energy
- attenuation** weaken or dilute
- atypical** irregular
- autoimmune** reaction of the immune system to the body's own tissues
- autoimmunity** immune reaction to the body's own tissues
- autosomal** describes a chromosome other than the X and Y sex-determining chromosomes
- autosome** a chromosome that is not sex-determining (not X or Y)
- axon** the long extension of a nerve cell down which information flows
- bacteriophage** virus that infects bacteria
- basal** lowest level



- base pair** two nucleotides (either DNA or RNA) linked by weak bonds
- basic** having the properties of a base; opposite of acidic
- benign** type of tumor that does not invade surrounding tissue
- binding protein** protein that binds to another molecule, usually either DNA or protein
- biodiversity** degree of variety of life
- bioinformatics** use of information technology to analyze biological data
- biolistic** firing a microscopic pellet into a biological sample (from biological/ballistic)
- biopolymers** biological molecules formed from similar smaller molecules, such as DNA or protein
- biopsy** removal of tissue sample for diagnosis
- biotechnology** production of useful products
- bipolar disorder** psychiatric disease characterized by alternating mania and depression
- blastocyst** early stage of embryonic development
- brackish** a mix of salt water and fresh water
- breeding analysis** analysis of the offspring ratios in breeding experiments
- buffers** substances that counteract rapid or wide pH changes in a solution
- Cajal** Ramon y Cajal, Spanish neuroanatomist
- carcinogens** substances that cause cancer
- carrier** a person with one copy of a gene for a recessive trait, who therefore does not express the trait
- catalyst** substance that speeds a reaction without being consumed (e.g., enzyme)
- catalytic** describes a substance that speeds a reaction without being consumed
- catalyze** aid in the reaction of
- cathode** negative pole
- cDNA** complementary DNA
- cell cycle** sequence of growth, replication and division that produces new cells
- centenarian** person who lives to age 100
- centromere** the region of the chromosome linking chromatids
- cerebrovascular** related to the blood vessels in the brain
- cerebrovascular disease** stroke, aneurysm, or other circulatory disorder affecting the brain

- charge density** ratio of net charge on the protein to its molecular mass
- chemotaxis** movement of a cell stimulated by a chemical attractant or repellent
- chemotherapeutic** use of chemicals to kill cancer cells
- chloroplast** the photosynthetic organelle of plants and algae
- chondrocyte** a cell that forms cartilage
- chromatid** a replicated chromosome before separation from its copy
- chromatin** complex of DNA, histones, and other proteins, making up chromosomes
- ciliated protozoa** single-celled organism possessing cilia, short hair-like extensions of the cell membrane
- circadian** relating to day or day length
- cleavage** hydrolysis
- cleave** split
- clinical trials** tests performed on human subjects
- codon** a sequence of three mRNA nucleotides coding for one amino acid
- Cold War** prolonged U.S.-Soviet rivalry following World War II
- colectomy** colon removal
- colon crypts** part of the large intestine
- complementary** matching opposite, like hand and glove
- conformation** three-dimensional shape
- congenital** from birth
- conjugation** a type of DNA exchange between bacteria
- cryo-electron microscope** electron microscope that integrates multiple images to form a three-dimensional model of the sample
- cryopreservation** use of very cold temperatures to preserve a sample
- cultivars** plant varieties resulting from selective breeding
- cytochemist** chemist specializing in cellular chemistry
- cytochemistry** cellular chemistry
- cytogenetics** study of chromosome structure and behavior
- cytologist** a scientist who studies cells
- cytokine** immune system signaling molecule
- cytokinesis** division of the cell's cytoplasm
- cytology** the study of cells
- cytoplasm** the material in a cell, excluding the nucleus



cytosol fluid portion of a cell, not including the organelles

de novo entirely new

deleterious harmful

dementia neurological illness characterized by impaired thought or awareness

demography aspects of population structure, including size, age distribution, growth, and other factors

denature destroy the structure of

deoxynucleotide building block of DNA

dimerize linkage of two subunits

dimorphism two forms

diploid possessing pairs of chromosomes, one member of each pair derived from each parent

disaccharide two sugar molecules linked together

dizygotic fraternal or nonidentical

DNA deoxyribonucleic acid

domains regions

dominant controlling the phenotype when one allele is present

dopamine brain signaling chemical

dosage compensation equalizing of expression level of X-chromosome genes between males and females, by silencing one X chromosome in females or amplifying expression in males

ecosystem an ecological community and its environment

ectopic expression expression of a gene in the wrong cells or tissues

electrical gradient chemiosmotic gradient

electrophoresis technique for separation of molecules based on size and charge

eluting exiting

embryogenesis development of the embryo from a fertilized egg

endangered in danger of extinction throughout all or a significant portion of a species' range

endogenous derived from inside the organism

endometriosis disorder of the endometrium, the lining of the uterus

endometrium uterine lining

endonuclease enzyme that cuts DNA or RNA within the chain

endoplasmic reticulum network of membranes within the cell

endoscope tool used to see within the body

endoscopic describes procedure wherein a tool is used to see within the body

endosymbiosis symbiosis in which one partner lives within the other

enzyme a protein that controls a reaction in a cell

epidemiologic the spread of diseases in a population

epidemiologists people who study the incidence and spread of diseases in a population

epidemiology study of incidence and spread of diseases in a population

epididymis tube above the testes for storage and maturation of sperm

epigenetic not involving DNA sequence change

epistasis suppression of a characteristic of one gene by the action of another gene

epithelial cells one of four tissue types found in the body, characterized by thin sheets and usually serving a protective or secretory function

Escherichia coli common bacterium of the human gut, used in research as a model organism

estrogen female hormone

et al. “and others”

ethicists a person who writes and speaks about ethical issues

etiology causation of disease, or the study of causation

eubacteria one of three domains of life, comprising most groups previously classified as bacteria

eugenics movement to “improve” the gene pool by selective breeding

eukaryote organism with cells possessing a nucleus

eukaryotic describing an organism that has cells containing nuclei

ex vivo outside a living organism

excise remove; cut out

excision removal

exogenous from outside

exon coding region of genes

exonuclease enzyme that cuts DNA or RNA at the end of a strand

expression analysis whole-cell analysis of gene expression (use of a gene to create its RNA or protein product)

fallopian tubes tubes through which eggs pass to the uterus

fermentation biochemical process of sugar breakdown without oxygen

- fibroblast** undifferentiated cell normally giving rise to connective tissue cells
- fluorophore** fluorescent molecule
- forensic** related to legal proceedings
- founder** population
- fractionated** purified by separation based on chemical or physical properties
- fraternal twins** dizygotic twins who share 50 percent of their genetic material
- frontal lobe** one part of the forward section of the brain, responsible for planning, abstraction, and aspects of personality
- gamete** reproductive cell, such as sperm or egg
- gastrulation** embryonic stage at which primitive gut is formed
- gel electrophoresis** technique for separation of molecules based on size and charge
- gene expression** use of a gene to create the corresponding protein
- genetic code** the relationship between RNA nucleotide triplets and the amino acids they cause to be added to a growing protein chain
- genetic drift** evolutionary mechanism, involving random change in gene frequencies
- genetic predisposition** increased risk of developing diseases
- genome** the total genetic material in a cell or organism
- genomics** the study of gene sequences
- genotype** set of genes present
- geothermal** related to heat sources within Earth
- germ cell** cell creating eggs or sperm
- germ-line** cells giving rise to eggs or sperm
- gigabase** one billion bases (of DNA)
- glucose** sugar
- glycolipid** molecule composed of sugar and fatty acid
- glycolysis** the breakdown of the six-carbon carbohydrates glucose and fructose
- glycoprotein** protein to which sugars are attached
- Golgi network** system in the cell for modifying, sorting, and delivering proteins
- gonads** testes or ovaries
- gradient** a difference in concentration between two regions
- Gram negative bacteria** bacteria that do not take up Gram stain, due to membrane structure



- Gram positive** able to take up Gram stain, used to classify bacteria
- gynecomastia** excessive breast development in males
- haploid** possessing only one copy of each chromosome
- haplotype** set of alleles or markers on a short chromosome segment
- hematopoiesis** formation of the blood
- hematopoietic** blood-forming
- heme** iron-containing nitrogenous compound found in hemoglobin
- hemolysis** breakdown of the blood cells
- hemolytic anemia** blood disorder characterized by destruction of red blood cells
- hemophiliacs** a person with hemophilia, a disorder of blood clotting
- herbivore** plant eater
- heritability** proportion of variability due to genes; ability to be inherited
- heritability estimates** how much of what is observed is due to genetic factors
- heritable** genetic
- heterochromatin** condensed portion of chromosomes
- heterozygote** an individual whose genetic information contains two different forms (alleles) of a particular gene
- heterozygous** characterized by possession of two different forms (alleles) of a particular gene
- high-throughput** rapid, with the capacity to analyze many samples in a short time
- histological** related to tissues
- histology** study of tissues
- histone** protein around which DNA winds in the chromosome
- homeostasis** maintenance of steady state within a living organism
- homologous** carrying similar genes
- homologues** chromosomes with corresponding genes that pair and exchange segments in meiosis
- homozygote** an individual whose genetic information contains two identical copies of a particular gene
- homozygous** containing two identical copies of a particular gene
- hormones** molecules released by one cell to influence another
- hybrid** combination of two different types
- hybridization (molecular)** base-pairing among DNAs or RNAs of different origins

- knocking out** deleting of a gene or obstructing gene expression
- laparoscope** surgical instrument that is inserted through a very small incision, usually guided by some type of imaging technique
- latent** present or potential, but not apparent
- lesion** damage
- ligand** a molecule that binds to a receptor or other molecule
- ligase** enzyme that repairs breaks in DNA
- ligate** join together
- linkage analysis** examination of co-inheritance of disease and DNA markers, used to locate disease genes
- lipid** fat or wax-like molecule, insoluble in water
- loci/locus** site(s) on a chromosome
- longitudinally** lengthwise
- lumen** the space within the tubes of the endoplasmic reticulum
- lymphocytes** white blood cells
- lyse** break apart
- lysis** breakage
- macromolecular** describes a large molecule, one composed of many similar parts
- macromolecule** large molecule such as a protein, a carbohydrate, or a nucleic acid
- macrophage** immune system cell that consumes foreign material and cellular debris
- malignancy** cancerous tissue
- malignant** cancerous; invasive tumor
- media** (bacteria) nutrient source
- meiosis** cell division that forms eggs or sperm
- melanocytes** pigmented cells
- meta-analysis** analysis of combined results from multiple clinical trials
- metabolism** chemical reactions within a cell
- metabolite** molecule involved in a metabolic pathway
- metaphase** stage in mitosis at which chromosomes are aligned along the cell equator
- metastasis** breaking away of cancerous cells from the initial tumor
- metastatic** cancerous cells broken away from the initial tumor
- methyrate** add a methyl group to



- methyated** a methyl group, CH_3 , added
- methylation** addition of a methyl group, CH_3
- microcephaly** reduced head size
- microliters** one thousandth of a milliliter
- micrometer** 1/1000 meter
- microsatellites** small repetitive DNA elements dispersed throughout the genome
- microtubule** protein strands within the cell, part of the cytoskeleton
- miscegenation** racial mixing
- mitochondria** energy-producing cell organelle
- mitogen** a substance that stimulates mitosis
- mitosis** separation of replicated chromosomes
- molecular hybridization** base-pairing among DNAs or RNAs of different origins
- molecular systematics** the analysis of DNA and other molecules to determine evolutionary relationships
- monoclonal antibodies** immune system proteins derived from a single B cell
- monomer** “single part”; monomers are joined to form a polymer
- monosomy** gamete that is missing a chromosome
- monozygotic** genetically identical
- morphologically** related to shape and form
- morphology** related to shape and form
- mRNA** messenger RNA
- muroid** having the properties of mucous
- mucosa** outer covering designed to secrete mucus, often found lining cavities and internal surfaces
- mucous membranes** nasal passages, gut lining, and other moist surfaces lining the body
- multimer** composed of many similar parts
- multinucleate** having many nuclei within a single cell membrane
- mutagen** any substance or agent capable of causing a change in the structure of DNA
- mutagenesis** creation of mutations
- mutation** change in DNA sequence
- nanometer** 10^{-9} (exp) meters; one billionth of a meter

- nascent** early-stage
- necrosis** cell death from injury or disease
- nematode** worm of the Nematoda phylum, many of which are parasitic
- neonatal** newborn
- neoplasms** new growths
- neuroimaging** techniques for making images of the brain
- neurological** related to brain function or disease
- neuron** nerve cell
- neurotransmitter** molecule released by one neuron to stimulate or inhibit a neuron or other cell
- non-polar** without charge separation; not soluble in water
- normal distribution** distribution of data that graphs as a bell-shaped curve
- Northern blot** a technique for separating RNA molecules by electrophoresis and then identifying a target fragment with a DNA probe
- Northern blotting** separating RNA molecules by electrophoresis and then identifying a target fragment with a DNA probe
- nuclear DNA** DNA contained in the cell nucleus on one of the 46 human chromosomes; distinct from DNA in the mitochondria
- nuclear membrane** membrane surrounding the nucleus
- nuclease** enzyme that cuts DNA or RNA
- nucleic acid** DNA or RNA
- nucleoid** region of the bacterial cell in which DNA is located
- nucleolus** portion of the nucleus in which ribosomes are made
- nucleoplasm** material in the nucleus
- nucleoside** building block of DNA or RNA, composed of a base and a sugar
- nucleoside triphosphate** building block of DNA or RNA, composed of a base and a sugar linked to three phosphates
- nucleosome** chromosome structural unit, consisting of DNA wrapped around histone proteins
- nucleotide** a building block of RNA or DNA
- ocular** related to the eye
- oncogene** gene that causes cancer
- oncogenesis** the formation of cancerous tumors
- oocyte** egg cell
- open reading frame** DNA sequence that can be translated into mRNA; from start sequence to stop sequence



opiate opium, morphine, and related compounds

organelle membrane-bound cell compartment

organic composed of carbon, or derived from living organisms; also, a type of agriculture stressing soil fertility and avoidance of synthetic pesticides and fertilizers

osmotic related to differences in concentrations of dissolved substances across a permeable membrane

ossification bone formation

osteoarthritis a degenerative disease causing inflammation of the joints

osteoporosis thinning of the bone structure

outcrossing fertilizing between two different plants

oviduct a tube that carries the eggs

ovulation release of eggs from the ovaries

ovules eggs

ovum egg

oxidation chemical process involving reaction with oxygen, or loss of electrons

oxidized reacted with oxygen

pandemic disease spread throughout an entire population

parasites organisms that live in, with, or on another organism

pathogen disease-causing organism

pathogenesis pathway leading to disease

pathogenic disease-causing

pathogenicity ability to cause disease

pathological altered or changed by disease

pathology disease process

pathophysiology disease process

patient advocate a person who safeguards patient rights or advances patient interests

PCR polymerase chain reaction, used to amplify DNA

pedigrees sets of related individuals, or the graphic representation of their relationships

peptide amino acid chain

peptide bond bond between two amino acids

percutaneous through the skin

phagocytic cell-eating

- phenotype** observable characteristics of an organism
- phenotypic** related to the observable characteristics of an organism
- pheromone** molecule released by one organism to influence another organism's behavior
- phosphate group** PO_4 group, whose presence or absence often regulates protein action
- phosphodiester bond** the link between two nucleotides in DNA or RNA
- phosphorylating** addition of phosphate group (PO_4)
- phosphorylation** addition of the phosphate group PO_4^{3-}
- phylogenetic** related to the evolutionary development of a species
- phylogeneticists** scientists who study the evolutionary development of a species
- phylogeny** the evolutionary development of a species
- plasma membrane** outer membrane of the cell
- plasmid** a small ring of DNA found in many bacteria
- plastid** plant cell organelle, including the chloroplast
- pleiotropy** genetic phenomenon in which alteration of one gene leads to many phenotypic effects
- point mutation** gain, loss, or change of one to several nucleotides in DNA
- polar** partially charged, and usually soluble in water
- pollen** male plant sexual organ
- polymer** molecule composed of many similar parts
- polymerase** enzyme complex that synthesizes DNA or RNA from individual nucleotides
- polymerization** linking together of similar parts to form a polymer
- polymerize** to link together similar parts to form a polymer
- polymers** molecules composed of many similar parts
- polymorphic** occurring in several forms
- polymorphism** DNA sequence variant
- polypeptide** chain of amino acids
- polyploidy** presence of multiple copies of the normal chromosome set
- population studies** collection and analysis of data from large numbers of people in a population, possibly including related individuals
- positional cloning** the use of polymorphic genetic markers ever closer to the unknown gene to track its inheritance in CF families
- posterior** rear

- prebiotic** before the origin of life
- precursor** a substance from which another is made
- prevalence** frequency of a disease or condition in a population
- primary sequence** the sequence of amino acids in a protein; also called primary structure
- primate** the animal order including humans, apes, and monkeys
- primer** short nucleotide sequence that helps begin DNA replication
- primordial soup** hypothesized prebiotic environment rich in life's building blocks
- probe** molecule used to locate another molecule
- procarcinogen** substance that can be converted into a carcinogen, or cancer-causing substance
- procreation** reproduction
- progeny** offspring
- prokaryote** a single-celled organism without a nucleus
- promoter** DNA sequence to which RNA polymerase binds to begin transcription
- promutagen** substance that, when altered, can cause mutations
- pronuclei** egg and sperm nuclei before they fuse during fertilization
- proprietary** exclusively owned; private
- proteomic** derived from the study of the full range of proteins expressed by a living cell
- proteomics** the study of the full range of proteins expressed by a living cell
- protists** single-celled organisms with cell nuclei
- protocol** laboratory procedure
- protonated** possessing excess H^+ ions; acidic
- pyrophosphate** free phosphate group in solution
- quiescent** non-dividing
- radiation** high energy particles or waves capable of damaging DNA, including X rays and gamma rays
- recessive** requiring the presence of two alleles to control the phenotype
- recombinant DNA** DNA formed by combining segments of DNA, usually from different types of organisms
- recombining** exchanging genetic material
- replication** duplication of DNA
- restriction enzyme** an enzyme that cuts DNA at a particular sequence

- retina** light-sensitive layer at the rear of the eye
- retroviruses** RNA-containing viruses whose genomes are copied into DNA by the enzyme reverse transcriptase
- reverse transcriptase** enzyme that copies RNA into DNA
- ribonuclease** enzyme that cuts RNA
- ribosome** protein-RNA complex at which protein synthesis occurs
- ribozyme** RNA-based catalyst
- RNA** ribonucleic acid
- RNA polymerase** enzyme complex that creates RNA from DNA template
- RNA triplets** sets of three nucleotides
- salinity** of, or relating to, salt
- sarcoma** a type of malignant (cancerous) tumor
- scanning electron microscope** microscope that produces images with depth by bouncing electrons off the surface of the sample
- sclerae** the “whites” of the eye
- scrapie** prion disease of sheep and goats
- segregation analysis** statistical test to determine pattern of inheritance for a trait
- senescence** a state in a cell in which it will not divide again, even in the presence of growth factors
- senile plaques** disease
- serum (pl. sera)** fluid portion of the blood
- sexual orientation** attraction to one sex or the other
- somatic** nonreproductive; not an egg or sperm
- Southern blot** a technique for separating DNA fragments by electrophoresis and then identifying a target fragment with a DNA probe
- Southern blotting** separating DNA fragments by electrophoresis and then identifying a target fragment with a DNA probe
- speciation** the creation of new species
- spindle** football-shaped structure that separates chromosomes in mitosis
- spindle fiber** protein chains that separate chromosomes during mitosis
- spliceosome** RNA-protein complex that removes introns from RNA transcripts
- spontaneous** non-inherited
- sporadic** caused by new mutations
- stem cell** cell capable of differentiating into multiple other cell types

- stigma** female plant sexual organ
- stop codon** RNA triplet that halts protein synthesis
- striatum** part of the midbrain
- subcutaneous** under the skin
- sugar** glucose
- supercoiling** coiling of the helix
- symbiont** organism that has a close relationship (symbiosis) with another
- symbiosis** a close relationship between two species in which at least one benefits
- symbiotic** describes a close relationship between two species in which at least one benefits
- synthesis** creation
- taxon/taxa** level(s) of classification, such as kingdom or phylum
- taxonomical** derived from the science that identifies and classifies plants and animals
- taxonomist** a scientist who identifies and classifies organisms
- telomere** chromosome tip
- template** a master copy
- tenets** generally accepted beliefs
- terabyte** a trillion bytes of data
- teratogenic** causing birth defects
- teratogens** substances that cause birth defects
- thermodynamics** process of energy transfers during reactions, or the study of these processes
- threatened** likely to become an endangered species
- topological** describes spatial relations, or the study of these relations
- topology** spatial relations, or the study of these relations
- toxicological** related to poisons and their effects
- transcript** RNA copy of a gene
- transcription** messenger RNA formation from a DNA sequence
- transcription factor** protein that increases the rate of transcription of a gene
- transduction** conversion of a signal of one type into another type
- transgene** gene introduced into an organism
- transgenics** transfer of genes from one organism into another
- translation** synthesis of protein using mRNA code

translocation movement of chromosome segment from one chromosome to another

transposable genetic element DNA sequence that can be copied and moved in the genome

transposon genetic element that moves within the genome

trilaminar three-layer

triploid possessing three sets of chromosomes

trisomics mutants with one extra chromosome

trisomy presence of three, instead of two, copies of a particular chromosome

tumor mass of undifferentiated cells; may become cancerous

tumor suppressor genes cell growths

tumors masses of undifferentiated cells; may become cancerous

vaccine protective antibodies

vacuole cell structure used for storage or related functions

van der Waal's forces weak attraction between two different molecules

vector carrier

vesicle membrane-bound sac

virion virus particle

wet lab laboratory devoted to experiments using solutions, cell cultures, and other “wet” substances

wild-type most common form of a trait in a population

Wilm's tumor a cancerous cell mass of the kidney

X ray crystallography use of X rays to determine the structure of a molecule

xenobiotic foreign biological molecule, especially a harmful one

zygote fertilized egg

Topic Outline

APPLICATIONS TO OTHER FIELDS

Agricultural Biotechnology
Biopesticides
Bioremediation
Biotechnology
Conservation Biology: Genetic Approaches
DNA Profiling
Genetically Modified Foods
Molecular Anthropology
Pharmacogenetics and Pharmacogenomics
Plant Genetic Engineer
Public Health, Genetic Techniques in
Transgenic Animals
Transgenic Microorganisms
Transgenic Plants

BACTERIAL GENETICS

Escherichia coli (*E. coli* bacterium)
Ames Test
Antibiotic Resistance
Chromosome, Prokaryotic
Cloning Genes
Conjugation
Eubacteria
Microbiologist
Overlapping Genes
Plasmid
Transduction
Transformation
Transgenic Microorganisms
Transgenic Organisms: Ethical Issues
Viroids and Virusoids
Virus

BASIC CONCEPTS

Biotechnology
Crossing Over

Disease, Genetics of
DNA
DNA Structure and Function, History
Fertilization
Gene
Genetic Code
Genetics
Genome
Genotype and Phenotype
Homology
Human Genome Project
Inheritance Patterns
Meiosis
Mendelian Genetics
Mitosis
Mutation
Nucleotide
Plasmid
Population Genetics
Proteins
Recombinant DNA
Replication
RNA
Transcription
Translation

BIOTECHNOLOGY

Agricultural Biotechnology
Biopesticides
Bioremediation
Biotechnology
Biotechnology and Genetic Engineering, History
Biotechnology: Ethical Issues
Cloning Genes
Cloning Organisms
DNA Vaccines



Genetically Modified Foods
HPLC: High-Performance Liquid Chromatography
Pharmaceutical Scientist
Plant Genetic Engineer
Polymerase Chain Reaction
Recombinant DNA
Restriction Enzymes
Reverse Transcriptase
Transgenic Animals
Transgenic Microorganisms
Transgenic Organisms: Ethical Issues
Transgenic Plants

CAREERS

Attorney
Bioinformatics
Clinical Geneticist
College Professor
Computational Biologist
Conservation Geneticist
Educator
Epidemiologist
Genetic Counselor
Geneticist
Genomics Industry
Information Systems Manager
Laboratory Technician
Microbiologist
Molecular Biologist
Pharmaceutical Scientist
Physician Scientist
Plant Genetic Engineer
Science Writer
Statistical Geneticist
Technical Writer

CELL CYCLE

Apoptosis
Balanced Polymorphism
Cell Cycle
Cell, Eukaryotic
Centromere
Chromosome, Eukaryotic
Chromosome, Prokaryotic
Crossing Over
DNA Polymerases
DNA Repair
Embryonic Stem Cells
Eubacteria
Inheritance, Extranuclear

Linkage and Recombination
Meiosis
Mitosis
Oncogenes
Operon
Polyploidy
Replication
Signal Transduction
Telomere
Tumor Suppressor Genes

CLONED OR TRANSGENIC ORGANISMS

Agricultural Biotechnology
Biopesticides
Biotechnology
Biotechnology: Ethical Issues
Cloning Organisms
Cloning: Ethical Issues
Gene Targeting
Model Organisms
Patenting Genes
Reproductive Technology
Reproductive Technology: Ethical Issues
Rodent Models
Transgenic Animals
Transgenic Microorganisms
Transgenic Organisms: Ethical Issues
Transgenic Plants

DEVELOPMENT, LIFE CYCLE, AND NORMAL HUMAN VARIATION

Aging and Life Span
Behavior
Blood Type
Color Vision
Development, Genetic Control of
Eye Color
Fertilization
Genotype and Phenotype
Hormonal Regulation
Immune System Genetics
Individual Genetic Variation
Intelligence
Mosaicism
Sex Determination
Sexual Orientation
Twins
X Chromosome
Y Chromosome

DNA, GENE AND CHROMOSOME STRUCTURE

Antisense Nucleotides
Centromere
Chromosomal Banding
Chromosome, Eukaryotic
Chromosome, Prokaryotic
Chromosomes, Artificial
DNA
DNA Repair
DNA Structure and Function, History
Evolution of Genes
Gene
Genome
Homology
Methylation
Multiple Alleles
Mutation
Nature of the Gene, History
Nomenclature
Nucleotide
Overlapping Genes
Plasmid
Polymorphisms
Pseudogenes
Repetitive DNA Elements
Telomere
Transposable Genetic Elements
X Chromosome
Y Chromosome

DNA TECHNOLOGY

In situ Hybridization
Antisense Nucleotides
Automated Sequencer
Blotting
Chromosomal Banding
Chromosomes, Artificial
Cloning Genes
Cycle Sequencing
DNA Footprinting
DNA Libraries
DNA Microarrays
DNA Profiling
Gel Electrophoresis
Gene Targeting
HPLC: High-Performance Liquid Chromatography
Marker Systems
Mass Spectrometry
Mutagenesis

Nucleases
Polymerase Chain Reaction
Protein Sequencing
Purification of DNA
Restriction Enzymes
Ribozyme
Sequencing DNA

ETHICAL, LEGAL, AND SOCIAL ISSUES

Attorney
Biotechnology and Genetic Engineering, History
Biotechnology: Ethical Issues
Cloning: Ethical Issues
DNA Profiling
Eugenics
Gene Therapy: Ethical Issues
Genetic Discrimination
Genetic Testing: Ethical Issues
Legal Issues
Patenting Genes
Privacy
Reproductive Technology: Ethical Issues
Transgenic Organisms: Ethical Issues

GENE DISCOVERY

Ames Test
Bioinformatics
Complex Traits
Gene and Environment
Gene Discovery
Gene Families
Genomics
Human Disease Genes, Identification of
Human Genome Project
Mapping

GENE EXPRESSION AND REGULATION

Alternative Splicing
Antisense Nucleotides
Chaperones
DNA Footprinting
Gene
Gene Expression: Overview of Control
Genetic Code
Hormonal Regulation
Imprinting
Methylation
Mosaicism
Nucleus

Operon
Post-translational Control
Proteins
Reading Frame
RNA
RNA Interference
RNA Polymerases
RNA Processing
Signal Transduction
Transcription
Transcription Factors
Translation

GENETIC DISORDERS

Accelerated Aging: Progeria
Addiction
Alzheimer's Disease
Androgen Insensitivity Syndrome
Attention Deficit Hyperactivity Disorder
Birth Defects
Breast Cancer
Cancer
Carcinogens
Cardiovascular Disease
Chromosomal Aberrations
Colon Cancer
Cystic Fibrosis
Diabetes
Disease, Genetics of
Down Syndrome
Fragile X Syndrome
Growth Disorders
Hemoglobinopathies
Hemophilia
Human Disease Genes, Identification of
Metabolic Disease
Mitochondrial Diseases
Muscular Dystrophy
Mutagen
Nondisjunction
Oncogenes
Psychiatric Disorders
Severe Combined Immune Deficiency
Tay-Sachs Disease
Triplet Repeat Disease
Tumor Suppressor Genes

GENETIC MEDICINE: DIAGNOSIS, TESTING, AND TREATMENT

Clinical Geneticist
DNA Vaccines

Embryonic Stem Cells
Epidemiologist
Gene Discovery
Gene Therapy
Gene Therapy: Ethical Issues
Genetic Counseling
Genetic Counselor
Genetic Testing
Genetic Testing: Ethical Issues
Geneticist
Genomic Medicine
Human Disease Genes, Identification of
Pharmacogenetics and Pharmacogenomics
Population Screening
Prenatal Diagnosis
Public Health, Genetic Techniques in
Reproductive Technology
Reproductive Technology: Ethical Issues
RNA Interference
Statistical Geneticist
Statistics
Transplantation

GENOMES

Chromosome, Eukaryotic
Chromosome, Prokaryotic
Evolution of Genes
Genome
Genomic Medicine
Genomics
Genomics Industry
Human Genome Project
Mitochondrial Genome
Mutation Rate
Nucleus
Polymorphisms
Repetitive DNA Elements
Transposable Genetic Elements
X Chromosome
Y Chromosome

GENOMICS, PROTEOMICS, AND BIOINFORMATICS

Bioinformatics
Combinatorial Chemistry
Computational Biologist
DNA Libraries
DNA Microarrays
Gene Families
Genome
Genomic Medicine

Genomics
 Genomics Industry
 High-Throughput Screening
 Human Genome Project
 Information Systems Manager
 Internet
 Mass Spectrometry
 Nucleus
 Protein Sequencing
 Proteins
 Proteomics
 Sequencing DNA

HISTORY

Biotechnology and Genetic Engineering, History
 Chromosomal Theory of Inheritance, History
 Crick, Francis
 Delbrück, Max
 DNA Structure and Function, History
 Eugenics
 Human Genome Project
 McClintock, Barbara
 McKusick, Victor
 Mendel, Gregor
 Morgan, Thomas Hunt
 Muller, Hermann
 Nature of the Gene, History
 Ribosome
 Sanger, Fred
 Watson, James

INHERITANCE

Chromosomal Theory of Inheritance, History
 Classical Hybrid Genetics
 Complex Traits
 Crossing Over
 Disease, Genetics of
 Epistasis
 Fertilization
 Gene and Environment
 Genotype and Phenotype
 Heterozygote Advantage
 Imprinting
 Inheritance Patterns
 Inheritance, Extranuclear
 Linkage and Recombination
 Mapping
 Meiosis
 Mendel, Gregor
 Mendelian Genetics

Mosaicism
 Multiple Alleles
 Nondisjunction
 Pedigree
 Pleiotropy
 Polyploidy
 Probability
 Quantitative Traits
 Sex Determination
 Twins
 X Chromosome
 Y Chromosome

MODEL ORGANISMS

Arabidopsis thaliana
Escherichia coli (*E. coli* Bacterium)
 Chromosomes, Artificial
 Cloning Organisms
 Embryonic Stem Cells
 Fruit Fly: *Drosophila*
 Gene Targeting
 Maize
 Model Organisms
 RNA Interference
 Rodent Models
 Roundworm: *Caenorhabditis elegans*
 Transgenic Animals
 Yeast
 Zebrafish

MUTATION

Chromosomal Aberrations
 DNA Repair
 Evolution of Genes
 Genetic Code
 Muller, Hermann
 Mutagen
 Mutagenesis
 Mutation
 Mutation Rate
 Nondisjunction
 Nucleases
 Polymorphisms
 Pseudogenes
 Reading Frame
 Repetitive DNA Elements
 Transposable Genetic Elements

ORGANISMS, CELL TYPES, VIRUSES

Arabidopsis thaliana
Escherichia coli (*E. coli* bacterium)

Archaea
 Cell, Eukaryotic
 Eubacteria
 Evolution, Molecular
 Fruit Fly: *Drosophila*
 HIV
 Maize
 Model Organisms
 Nucleus
 Prion
 Retrovirus
 Rodent Models
 Roundworm: *Caenorhabditis elegans*
 Signal Transduction
 Viroids and Virusoids
 Virus
 Yeast
 Zebrafish

POPULATION GENETICS AND EVOLUTION

Antibiotic Resistance
 Balanced Polymorphism
 Conservation Biologist
 Conservation Biology: Genetic Approaches
 Evolution of Genes
 Evolution, Molecular
 Founder Effect
 Gene Flow
 Genetic Drift
 Hardy-Weinberg Equilibrium

Heterozygote Advantage
 Inbreeding
 Individual Genetic Variation
 Molecular Anthropology
 Population Bottleneck
 Population Genetics
 Population Screening
 Selection
 Speciation

RNA

Antisense Nucleotides
 Blotting
 DNA Libraries
 Genetic Code
 HIV
 Nucleases
 Nucleotide
 Reading Frame
 Retrovirus
 Reverse Transcriptase
 Ribosome
 Ribozyme
 RNA
 RNA Interference
 RNA Polymerases
 RNA Processing
 Transcription
 Translation